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Contact Monitor:		**Contact Phone:**	
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	The lymphocyte-based adaptive immune system is a complex cellular structure which seems to have evolved entirely since the vertebrate line diverged from all other deuterostome animals. The long-term goal of this project is to discover the molecular basis for the many interlinked biological innovations that made this system possible. In the first period of funding of this project, we have made swift progress in identifying the lower-vertebrate and related invertebrate versions of the major regulatory genes that control lymphocyte development, the genes encoding a key set of hematopoietic transcription factors. This work has revealed shifts in the number, primary structure, and expression patterns of corresponding genes, with major changes apparently between the divergence times of jawless vertebrates and cartilaginous vertebrates. In this renewal proposal, we will focus on three areas. The first is the comparison between the skate, which we have found to use a full set of lymphoid regulatory genes in a similar way to mammals, and the		

Task Description:	lamprey, in which initial studies of two genes indicate substantial differences in structure and/or gene number from all jawed vertebrates. We will characterize the lamprey homologues of these transcription factor genes to test the timing of gene duplications. We will determine whether conserved DNA binding domains may be linked with different protein interaction domains. Expression studies will be done to try to explain why lampreys have no true lymphocytes. Combinations of co-expressed factors in lamprey tissues will be sought as evidence for possible cryptic lymphocyte-like cells in these animals. A major new aim of this project has recently been made possible by the discovery of the genes encoding the immune receptors of lampreys by our collaborators, Zeev Pancer and Max Cooper, in 2004. Fascinatingly, these receptors are completely different in structure from the familiar immunoglobulins and T-cell receptors used by immune cells in jawed vertebrates; however, they appear to carry out similar effector functions. Therefore, this gene complex must be the direct target of any transcription factors that are used by lamprey lymphocyte-like cells to program them for antigen recognition. Using the genomic DNA clones that encode these receptors, we will therefore map the sequences that are critical for the expression of the lamprey receptors by transfection studies, and then use this information to zero in on the most important transcription factors. This task is to reveal the evolutionary mechanism that enabled the ancestors of vertebrates to develop a unique new pair of cell types, the T and B lymphocytes of the immune system. This task is important to understand the ways that novel body parts and cell types appear in evolution.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work is to obtain a deeper understanding of molecular mechanisms driving the evolutionary process by focusing on two very well-understood cell types that are known to have evolved only in vertebrates. The processes that enabled these cell types to be generated are a model for the way organisms create new structures generally. A better understanding of mechanisms giving life forms adaptibility is a benefit.
Task Progress:	Task progress for 2006 Period covered: 10/2004-9/2006 (no-cost extension) In the final (extended) year of this project, we capitalized on the availability of a new probe for the evolutionary relationships between jawed vertebrate lymphocytes and the cells that perform analogous functions in other organisms. We obtained the genomic DNA clones encoding the basic structure of the lamprey VLR locus, i.e. the gene family that is used by lampreys in place of the T-cell receptor and immunoglobulin genes of jawed vertebrates. Because of the structures of the locus and its lymphocyte-specific rearrangements, we reasoned that the regulatory sequences for tissue-specific control must lie in an interval of approximately 12 kb that was contained in two of the genomic clones we received. Thus, the regulatory uniqueness of lamprey lymphocytes could be reflected in the constellation of regulatory factors needed to control this VLR gene through binding to sites in this ~12 kb region. This approach converges with the approach we originally pursued. That approach was to identify cell types by their characteristic combinations of regulatory molecules, and to establish homology or nonhomology between lamprey and higher vertebrate lymphocytes through the comparison. In a regulatory sense, this is a “top down” approach. We started by identifying and tracking lamprey transcription factors which had been chosen by homology to those used in higher vertebrate lymphocytes, and then sought to assess whether or not the same spectrum of factors are present in the lamprey lymphocytes. We were successful in finding probes to measure expression of a variety of lamprey transcription factors homologous to vertebrate lymphocyte factors. We then purified lamprey lymphocytes and determined the expression pattern of these factors in lymphocytes as opposed to other tissues, using a series of recently described lamprey lymphocyte markers to compare enrichment of these factors. Ideally, among the transcription factors we studied there should be the regulators of the lamprey VLR locus, if indeed these transcription factors are used in lampreys in homologous ways to their use in jawed vertebrate lymphocytes. Our results provided some support for conserved use of a common set of transcription factors. GATA-2/3 and IKLF1 (homologs of T-lineage-associated transcription factors in jawed vertebrates) were particularly strongly enriched in the lymphocyte-like population, along with the markers “CD4-like” and “TCR-like”. A trend toward enrichment in lymphocytes and lymphocyte-rich tissues was also seen with lamprey Spi and lamprey Pax2/5/8 (homologs of B lineage-associated transcription factors). However, the expression levels of these transcription factors did not show as dramatic an enrichment in the lymphocyte populations as the lymphocyte marker genes, and the scatter in the data was high. This could be because of dilution caused by heterogeneity in the lymphocyte population, but it could also be because the factors actually used by lamprey lymphocytes are different, in fact, from those used in higher vertebrates. Thus a complementary approach was needed. By starting from a vitally important gene complex known to function specifically in lamprey lymphocytes, we could identify all the classes of transcription factors involved and then determine their homology or lack of homology to those of lymphocytes in jawed vertebrates. Dr. Jonathan Moore joined this project from a lab known for its expertise in the bioinformatic analysis of evolutionary relationships. He carried out an alignment and sequence motif analysis on two dissimilar genomic clones from the VLR locus, and identified the highly conserved blocks within the 12-15 kb region within which the regulatory sequences could be deduced to reside. This region was then scanned for potential transcription factor binding sites based on MatInspector position weight matrices, and three regions of interest with intense clustering of predicted sites were selected for closer study. Dr. Moore then set up a heterologous transfection system to test different lamprey VLR flanking sequences for their activity in driving reporter expression in reference immune cell lines of jawed vertebrates. No tissue culture system is available to test gene expression in lamprey lymphocytes themselves, but the rationale for our strategy was that if a particular set of transcription factors did exist that exerted a conserved function in jawless and jawed vertebrate immune cells, then we might detect a cell-type specific increase in transcription from the sequences recognized by these factors, even in heterologous cells. The results of his scan of the ~12 kb region revealed that none of these sequences conferred any particular function in murine T cells or murine macrophages. However, a particular region upstream of the VLR promoter sharply enhanced the expression of a reporter if it was assayed in murine B cells. This result was reproduced with two independent sets of DNA, and further testing has helped to narrow down the region that may be most critical for the enhanced activity in the context of mammalian B-cell transcription factors. This provides the “hook” with which one can now “fish” for regulatory factors that may exert conserved functions in the mammalian and jawless vertebrate immune cells in spite of the vast evolutionary change in modes of immune recognition that have occurred. We are very anxious to identify these factors now and test directly for whether they are the same in mammals and lampreys.

	As this project has ended and the NASA program that supported it has also ended, we hope to use the exciting results of this analysis to seek further support from other agencies, such as NSF. Dr. Jonathan Moore has been awarded an NIH NRSA for this work, which is helping him to continue the project even though NASA funding has ceased. While the highly exploratory nature of this most recent research made it impossible to bring it to publication within the time frame of this grant, we fully expect the work to be published in the next few years.
Bibliography Type:	Description: (Last Updated: 03/17/2017)
Articles in Peer-reviewed Journals	Rothenberg EV, Pant R. "Origins of lymphocyte developmental programs: transcription factor evidence." Semin Immunol. 2004 Aug;16(4):227-38. Review. PMID: 15522621 , Aug-2004
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